

After Deep CFR: four modern poker AIs

Pluribus, ReBeL, Player of Games, AlphaHoldem — a pedagogical tour

Companion volume to *Deep CFR, explained*

PokerTrainer documentation

June 2026

Abstract

The companion tutorial *Deep CFR, explained* covered the theory line from regret matching to Deep CFR. This volume takes the next step and dissects four landmark systems that followed: **Pluribus** (superhuman six-player poker with *no* neural network), **ReBeL** (reinforcement learning + search on *belief states*), **Player of Games** (one sound algorithm for chess, Go *and* poker), and **AlphaHoldem** (end-to-end deep RL, no search at all). For each: the one core idea, how it works mechanically (diagrams + Python pseudo-code), what it achieved, and what it costs. A final section compares all of them — including Deep CFR — on one table and draws lessons for this project.

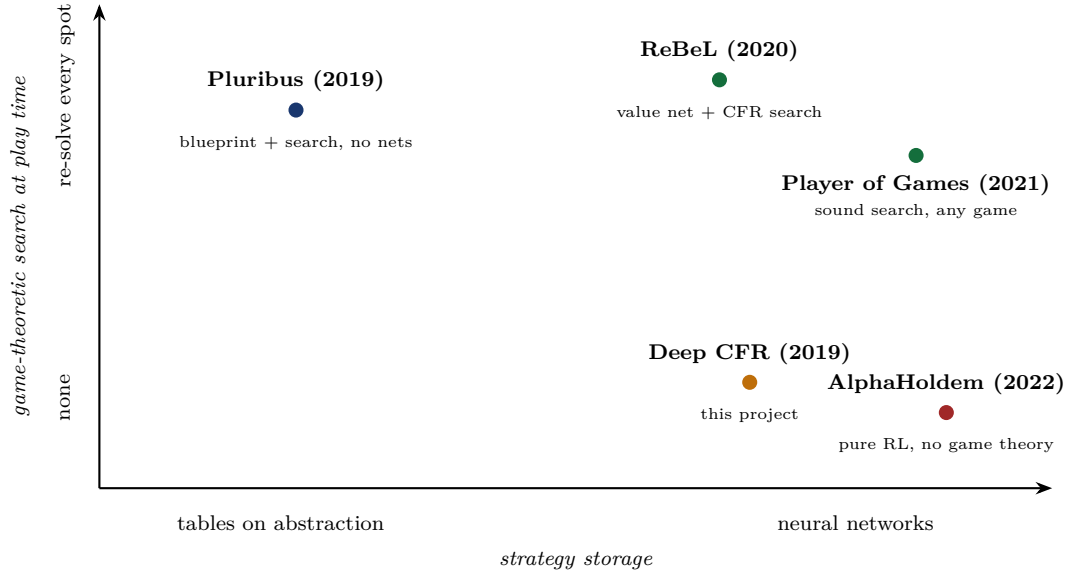
Contents

1	The design space: one map before four deep dives	3
1.1	Refresher: the CFR core that three of the four systems share	3
2	Pluribus: superhuman six-player poker, no neural network	4
2.1	Why six players changes the rules of the game	4
2.2	Stage 1 (offline): the blueprint	4
2.3	Stage 2 (online): re-solving — but what exactly do you re-solve?	4
2.4	Trick 1: depth limits with continuation strategies	5
2.5	Trick 2: solve for your whole range, not for your hand	6
2.6	The full play-time loop	6
2.7	Close-up: the two questions everyone asks	7
3	ReBeL: reinforcement learning + search on belief states	8
3.1	The conversion trick	9
3.2	The self-play loop, one hand at a time	9
3.3	One solve under the microscope: a worked miniature	11
3.4	Two phrases from the listing, unpacked	12
3.5	What the value net actually sees and says	14
3.6	Close-up: the two questions everyone asks	14
4	Player of Games: one sound algorithm for chess, Go and poker	15
4.1	What “sound” buys you	15

4.2	One GT-CFR round under the microscope: growing a tree	16
4.3	What a node of the search tree actually is	17
4.4	What the CVPN actually sees and says	18
4.5	Two phrases unpacked	18
4.6	Close-up: the two questions everyone asks	19
5	AlphaHoldem: what if we drop the game theory entirely?	20
5.1	Architecture: two eyes, one brain	20
5.2	One observation under the microscope: the two tensors	21
5.3	Making PPO survive poker	21
5.4	K-best self-play against strategy cycling	22
5.5	The cycle under the microscope: rock–paper–scissors in strategy space	22
5.6	Two phrases unpacked	23
5.7	Close-up: the two questions everyone asks	24
6	Side by side, and what this project takes from each	25
7	Further reading	25

1 The design space: one map before four deep dives

Every poker AI answers two independent questions. (1) **Where does game-theoretic reasoning happen?** Offline only (train a strategy, then just play it), or also *online* (re-solve the current situation at the table)? (2) **What stores the strategy?** Tables over abstracted infosets, or neural networks? The four systems in this volume occupy four distinct corners:



Intuition

Read the map as a tug-of-war between two resources: **offline training compute** (bottom row pays it all upfront, then plays instantly) and **online thinking time** (top row keeps solving at the table, which buys precision but costs seconds and engineering). The diagonal from Pluribus to AlphaHoldem is also a diagonal of *decreasing game theory and increasing deep learning*.

1.1 Refresher: the CFR core that three of the four systems share

Pluribus fills its blueprint with it, ReBeL runs it inside every subgame, Player of Games runs its regret-update phase with it — only AlphaHoldem drops it. So here is tabular CFR once, compactly, with its two classic upgrades shown inline (the full pedagogical build-up lives in the companion volume, Sections 2–3):

Listing 1: CFR in one block. `reach[p]` is the probability that player p 's own choices lead here; regrets are weighted by the *opponent's* reach, the average strategy by *your own*. The commented lines show the vanilla / CFR⁺ variants of the update; the active lines are Linear CFR (weight everything by t), which is what Pluribus and Deep CFR use.

```
R, S = tables_of_zeros(), tables_of_zeros() # regrets / average strategy

def cfr(state, t, reach):
    if state.is_terminal():
        return state.payoffs() # (u_0, u_1), zero-sum
    I, p = infoset(state), state.to_act()
    sigma = regret_matching(R[I]) # positive part, normalized
    v, v_I = {}, [0.0, 0.0]
    for a in state.legal():
        r = list(reach)
        r[p] *= sigma[a] # my move scales MY reach
```

```

    v[a] = cfr(state.child(a), t, r)
    v_I = [x + sigma[a] * y for x, y in zip(v_I, v[a])]
    for a in state.legal():
        # vanilla CFR: R[I][a] += reach[1-p] * (v[a][p] - v_I[p])
        # CFR+: R[I][a] = max(R[I][a] + reach[1-p] * (.), 0)
        R[I][a] += t * reach[1 - p] * (v[a][p] - v_I[p]) # Linear CFR
    S[I] += t * reach[p] * sigma # the average strategy = the output
    return v_I

def solve(T):
    for t in range(1, T + 1):
        cfr(deal_random_hand(), t, reach=[1.0, 1.0]) # chance sampled
    return normalize(S) # converges to Nash (2 players)

```

2 Pluribus: superhuman six-player poker, no neural network

Paper: Brown & Sandholm, *Superhuman AI for multiplayer poker*, Science 2019. **One idea:** a coarse offline “blueprint” strategy plus cheap *depth-limited* online re-solving is enough — even with six players, even with tables instead of networks.

2.1 Why six players changes the rules of the game

Everything in the Deep CFR tutorial relied on the two-player zero-sum correspondence: regret minimization $\Rightarrow$ Nash, and Nash $\Rightarrow$ unbeatable. With six players *both* halves break:

In games with 3+ players, computing a Nash equilibrium is computationally intractable, and — worse — *playing* one is no longer a guarantee: if the others deviate, your equilibrium share can lose. Pluribus therefore drops the theoretical target entirely and aims for an *empirical* one: a strategy that consistently beats elite humans. CFR-style self-play is kept as a *heuristic* that happens to produce strong, balanced play.

2.2 Stage 1 (offline): the blueprint

The blueprint is classic pre-deep-learning machinery, exactly what Deep CFR was designed to replace — and Pluribus shows it is still viable when paired with search:

- **Information abstraction:** strategically similar hands are bucketed (k-means on equity distributions), shrinking 10^{161} situations to a tractable table.
- **Action abstraction:** only a handful of bet sizes exist in the abstract game.
- **Linear MCCFR** (the same external-sampling traversals as our trainer, plus the same linear t -weighting) fills the regret tables — this is exactly the Linear-CFR update of Listing 1, run with sampling.

The famous headline: the blueprint trained in **8 days on a 64-core server**, under 512 GB RAM, **no GPU** — roughly \$150 of cloud compute, versus millions of dollars of TPU time for Go-class systems.

2.3 Stage 2 (online): re-solving — but what exactly do you re-solve?

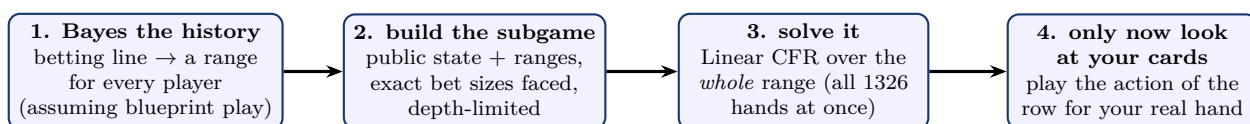
The blueprint alone is too coarse to beat pros (the abstraction blurs exact bet sizes and hand distinctions), so at the table Pluribus computes a fresh, finer strategy for the situation it is actually

in. In chess that sentence is easy to make precise: *search from the current position*. In poker the analogous sentence does not even type-check, and seeing why is most of understanding online re-solving.

Problem 1: the “position” is not a state. What the table shows (board, pot, betting history) is only the *public state*; its value depends on which hands the players might be holding. So the root of a poker subgame must carry, for every player, a *range*: a probability distribution over all 1326 hole-card pairs. And ranges do not fall from the sky — they come from strategies. *If* you assume each player has been following a known strategy so far (the blueprint, plus the outputs of any earlier re-solves), then every public action they took is evidence about their hand, and Bayes’ rule turns the betting history into a range: hands that would have folded to the flop raise get probability ~ 0 , hands that would have 3-bet get boosted, and so on. Pluribus knows its *own* strategy exactly, so its own range is exact; for opponents it assumes they played approximately the blueprint.

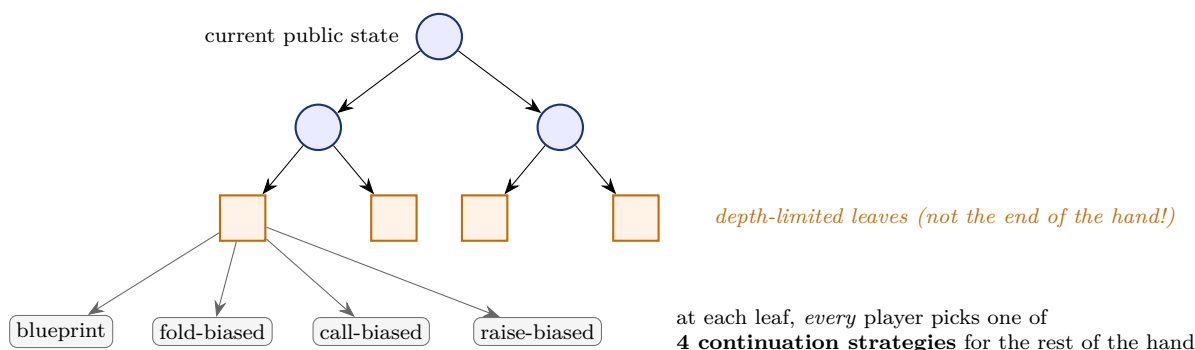
Problem 2: the subgame still reaches too deep. Even rooted on the flop, a no-limit subgame spans turn and river — far too large to solve in seconds. Truncating it at a depth limit needs care, and that is the paper’s key invention.

So one online solve looks like this end to end:



2.4 Trick 1: depth limits with continuation strategies

Cutting the subgame off at a depth limit means writing a *number* at each cut point: “the value of arriving here”. Any *single* assumption about how play continues below the cut (e.g. “everyone plays the blueprint afterwards”) is an assumption the search above will learn to exploit. The fix:



Intuition

Why continuation strategies make shallow leaves honest. If you evaluate a leaf with a *single* fixed policy (say, the blueprint), the search can “cheat”: it steers play toward leaves where the blueprint plays badly for the opponent, producing a strategy that only works if the opponent cooperates. Letting every player *choose among several continuations* at the leaf is a miniature game: the value of the leaf is what you can get when the opponent picks their best continuation *against you*. The search can no longer assume a docile opponent — at a tiny fraction of the cost of solving to the end of the hand.

The three biased continuations are cheap to build — take the blueprint, multiply the probability of the biased action (fold / call / raise) by a constant, renormalize — and the value of a leaf under any combination of choices is estimated by rolling the hand out under those strategies. The choices themselves are folded into the CFR solve: each player’s leaf-choice is just one more decision in the

subgame, so the equilibrium the solver finds is already robust against all 4^{players} ways play could continue.

2.5 Trick 2: solve for your whole range, not for your hand

A chess engine searches for the best move *in the position it sees*. The poker analogue — “find the best line for the two kings I am holding” — would be a disaster, and understanding why is the other half of online re-solving. Your opponents never observe your hand; they observe your *actions*. If the strategy you compute is a function of your actual hand, your actions become a code for it, and an attentive opponent reads the code: you bet only your strong hands, they fold exactly then, your strong hands stop earning. The whole point of a balanced (GTO) strategy is that the *same* observable action is taken by strong and weak hands in calibrated proportions.

So step 3 of the pipeline solves the subgame **for every hand Pluribus could be holding at once**: the output σ is a whole table — one row per hole-card pair in its range, one action distribution per row — computed jointly so that the *range as a whole* is unexploitable. Only after the solve does Pluribus look at its actual cards and read off that single row (step 4). Bluffs are not a separate mechanism: the equilibrium simply assigns some raising probability to weak rows so that strong rows get paid — balance is an *output* of range-level solving, not a style knob.

Intuition

This also explains an implementation detail: “vector-based” Linear CFR. Since the subgame must be solved for all 1326 hands anyway, the efficient form sweeps the public tree once per iteration carrying a 1326-long *vector* of regrets/values per node, instead of traversing once per hand. Small subgames use this exact vector form; very large ones fall back to sampling (Monte Carlo Linear CFR).

2.6 The full play-time loop

Putting both tricks into the pipeline, here is what happens at a Pluribus decision on, say, the flop:

1. **Freeze the public state**: board, pot, the exact bet it faces (no rounding — the subgame is built with the true size).
2. **Compute ranges** for all players by Bayes from the betting history, assuming blueprint play (its own range: exact).
3. **Build the depth-limited subgame** from this root — typically the rest of the current street plus the start of the next — and attach the four continuation choices at every leaf (Trick 1).
4. **Solve with Linear CFR for the whole range** (Trick 2), a few seconds on two CPUs.
5. **Play**: sample an action from the solved row of its actual hand.
6. **If an opponent later uses a bet size outside the current tree**: build a *new* subgame that contains that exact size and re-solve — *nested* re-solving, repeated as often as needed. (This is how off-tree actions are handled exactly instead of being rounded to the abstraction, the classic leak of blueprint-only bots.)

Listing 2: Pluribus, schematically. The blueprint is Deep CFR’s ancestor (same traversals, tables instead of nets); play time adds range-aware, depth-limited re-solving.

```
def blueprint(T):                                # OFFLINE, once, ~8 days on CPUs
    for t in range(1, T + 1):
        for p in range(6):                        # six players!
            mccfr_traverse(deal(), p, t)          # external sampling, linear weights
```

```

return regret_tables                # indexed by ABSTRACT infosets

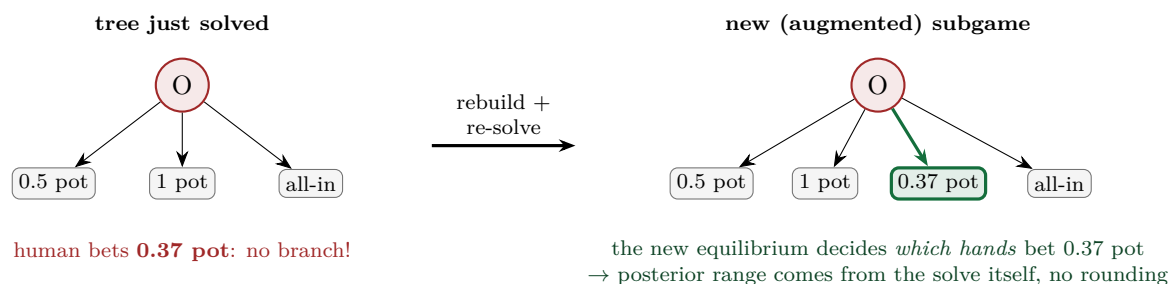
def resolve(public_state, ranges):    # ONLINE: one subgame solve
    G = subgame(public_state, ranges, # root carries the RANGES
                depth_limit=rest_of_street_plus,
                bet_sizes=exact_bets_faced) # no rounding
    for leaf in G.leaves():          # Trick 1: 4-way continuation choice
        leaf.attach_choices([blueprint, fold_bias, call_bias, raise_bias])
    return linear_cfr(G)             # Trick 2: sigma[hand] for ALL hands

def act(history, board, my_hand):
    if preflop and on_tree(history): # preflop abstraction is dense:
        return sample(blueprint[abstract(history, my_hand)])
    ranges = bayes_ranges(history, board, assumed=blueprint)
    sigma = resolve(public_state(history, board), ranges)
    return sample(sigma[my_hand])     # look at MY cards only at the end
# off-tree opponent bet later? build a new subgame with that exact size
# and call resolve() again (nested re-solving).

```

2.7 Close-up: the two questions everyone asks

Q1 — an opponent bets a size that is not in the tree. What happens, mechanically? Suppose the subgame Pluribus just solved offered the opponent three sizes at some node — {0.5 pot, 1 pot, all-in} — and the human bets 0.37 pot. The solution Pluribus is holding has *no entry* for what just happened. The historical fix (“action translation”: round 0.37 to the nearest size and pretend) is exactly the leak humans learned to attack by betting *between* the abstraction points. Pluribus instead rebuilds:



1. **Rebuild:** a fresh subgame rooted just before the bet, with the opponent’s action set at that node *augmented* by the exact 0.37-pot size.
2. **Ranges at the new root** are taken from *before* the off-tree action — everything up to that moment was on-model, so Bayes is still well-defined.
3. **Re-solve** the augmented subgame as usual (depth limit, continuation leaves, whole-range Linear CFR).
4. The subtle part: the new equilibrium contains **a strategy for the opponent at the augmented node** — it computes *which hands would bet 0.37 pot at equilibrium*. Bayes against *that* computed strategy gives the range going forward. The off-tree action is never mapped to a fiction; the solver *invents its correct interpretation* as part of the equilibrium.
5. Another off-tree size on a later street? Do it all again — hence *nested* re-solving.

Intuition

The asymmetry to remember: Pluribus’s *own* bet sizes can stay few — it chooses them. It is the *opponent’s* freedom that makes the full game impossible to pre-solve. Nested re-solving prices each opponent choice *at the moment it happens* instead of pre-computing responses to a continuum — the same “pay compute only for situations that actually arise” logic that motivates search in the first place. (Preflop, where Pluribus plays the blueprint directly, the action abstraction is dense enough that rounding a rare off-tree size costs little.)

Q2 — the ranges assume opponents follow the blueprint. What if they just... don’t?
Separate three cases.

Case 1: the deviation is an off-tree action. That is Q1 — structural, fully handled; the augmented equilibrium even defines what the new action “means”.

Case 2: on-tree actions, wrong frequencies. The human raises twice as often as the blueprint would. Pluribus’s posterior range for them is then simply *wrong* — it credits the line with blueprint frequencies. And Pluribus does *nothing* about it: by deliberate design it has no opponent modeling and never adapts to observed tendencies. Partly robustness (an adapting bot can be manipulated by feigned tendencies for later betrayal), partly principle: its defense is **balance, not mind-reading**. A range is a statement about a *model*, not about the human — Bayes answers “which hands would have taken this line *if* the opponent played the assumed strategy”.

Case 3: so why is playing on-model still sound? Zero-sum equilibrium logic: the strategy Pluribus computes is (approximately) unexploitable *within the model game*, and an opponent playing off-model frequencies is, from the model’s viewpoint, deviating from equilibrium — deviations do not gain in expectation. The model’s mixed strategies also keep ranges from ever collapsing to zero on the human’s true holding (regret matching leaves small probability on many actions), and every nested re-solve refreshes the range model. The honest catch is the next box.

Watch out

Where the soundness leak is. An opponent can deviate *early*, accept a small on-model loss, *poison the assumed range*, and harvest a larger error from the re-solver later — this is exactly what makes naive (“unsafe”) re-solving theoretically attackable in two-player zero-sum. Libratus closed the hole with *safe* re-solving gadgets that guarantee the re-solved strategy never does worse than the blueprint against any opponent hand. Pluribus — playing six-max, where equilibrium guarantees were abandoned anyway (Sec. 2.1) — accepts the heuristic; empirically, five elite professionals over 10 000 hands did not find a way in. One more instance of the multiplayer trade: theory exchanged for scale, validated at the table.

Results. Against elite professionals (including World Series champions), in both 5 humans + 1 AI and 1 human + 5 copies formats, Pluribus won decisively. Each decision takes seconds on two CPUs.

Watch out

Pluribus is often summarized as “deep learning beats poker pros”. It contains **no neural network at all**. Its lesson is the opposite of the hype: *search and good abstractions are devastatingly cost-effective*. When you read a new-method paper, always ask what a tuned blueprint-plus-search baseline would have done.

3 ReBeL: reinforcement learning + search on belief states

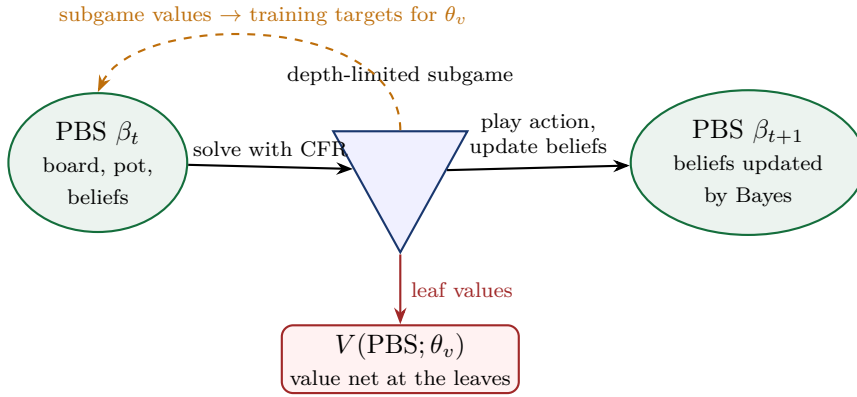
Paper: Brown, Bakhtin, Lerer, Gong, *Combining Deep Reinforcement Learning and Search for Imperfect-Information Games*, NeurIPS 2020. **One idea:** an imperfect-information game becomes a *perfect-information* game if the “state” is taken to be the **public belief state** — then

AlphaZero-style self-play + search applies, with CFR doing the searching.

3.1 The conversion trick

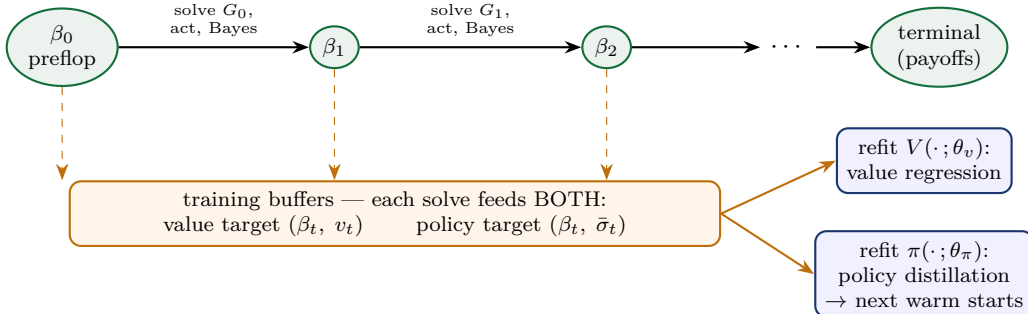
Public belief state (PBS). Everything publicly visible (board, pot, betting history) *plus*, for each player, the probability distribution over their possible private hands, as computable by an outside observer who knows both players' strategies. In hold'em each belief is a vector over the 1326 possible hole-card pairs.

Why this works: if both players' policies are common knowledge, then Bayes' rule updates the beliefs after every public action — so the PBS evolves *deterministically given the policies*, like a board position. Hidden information has been traded for a bigger, continuous state. Values become well-defined functions of the PBS, so a neural network $V(\text{PBS})$ can be trained the way AlphaZero trains its value net.



3.2 The self-play loop, one hand at a time

The loop is easiest to follow as a single hand of self-play, narrated. First, the cast: ReBeL plays *both* seats. At the start of an episode a real deal hands private cards to each seat (the *world state*); on top of that, the algorithm maintains one shared bookkeeping object, the PBS — which contains no secrets (board, pot, two belief vectors), so it looks identical from both seats. The hand then alternates between *solving* (a fresh subgame at the current PBS) and *stepping* (an action is played, beliefs update, possibly new board cards arrive):



At each decision point, six things happen, in order:

1. **Freeze the current PBS β_t :** board, pot, and both belief vectors as they stand.
2. **Build the depth-limited subgame G_t** rooted at β_t — roughly the rest of the current betting round, stopping wherever new cards would be dealt. Building fixes the *shape only*: no value is

attached to any leaf yet, because a leaf’s value depends on the beliefs *at that leaf*, and those do not exist until a strategy says how players reach it — which is exactly what the next step is about to compute.

3. **Solve it:** T_{cfr} iterations of plain *tabular* CFR inside G_t (volume 1’s algorithm, literally), with leaf evaluation happening *inside every iteration*: the current iterate’s σ induces beliefs at each leaf (Bayes along the path), the net is queried at that freshly built leaf PBS, and the returned per-hand values feed the regret updates. Strategy and leaf values *co-evolve* — every iteration re-prices the leaves under the new beliefs. The policy net θ_π warm-starts the iterations so fewer are needed.
4. **Store two training pairs** — one solve feeds both networks. Let $\bar{\sigma}_t$ be the *average* of the T_{cfr} policies (the object volume 1 proved converges). For the **value net**: (β_t, v_t) , where v_t is the per-hand value vector of the root under $\bar{\sigma}_t$ — “what this PBS is worth under near-equilibrium play”, AlphaZero-style value labels. For the **policy net**: $(\beta_t, \bar{\sigma}_t)$ itself, the root’s per-hand action distributions — distilled later by θ_π , these become tomorrow’s warm starts.
5. **Choose what to actually play:** pick *one random iteration* $t' \sim \text{uniform}(1..T_{\text{cfr}})$ and use its policy $\sigma_{t'}$ — *not* the average. This is the strangest-looking choice in the method; both reasons (training-distribution consistency and safety-by-randomization) are unpacked in the close-up, Q2 below.
6. **Act and update:** the seat to act looks at its real cards and samples its row of $\sigma_{t'}$; the action is public; Bayes’ rule pushes *both* belief vectors through $\sigma_{t'}$, giving β_{t+1} (if a street ends, the dealt board cards update the PBS too).

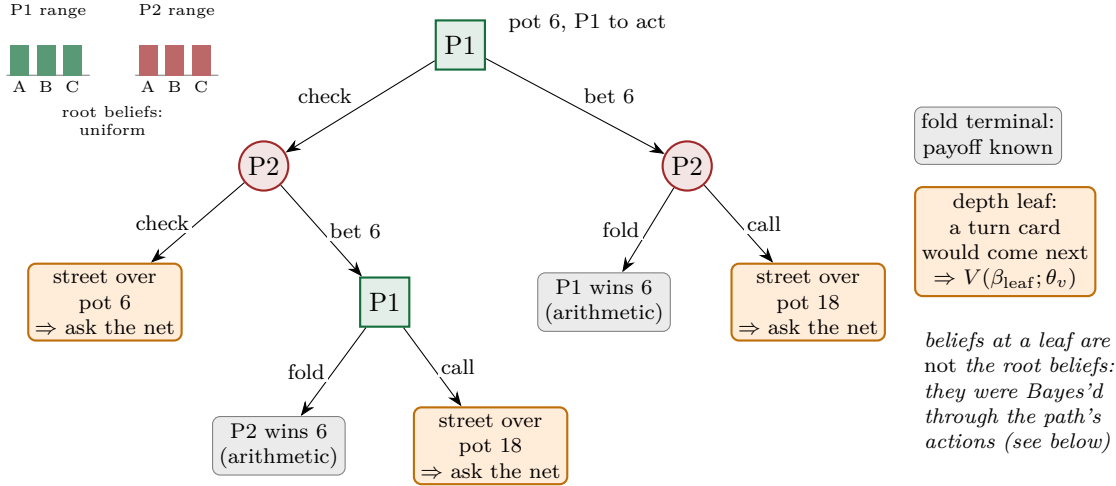
When the hand reaches a terminal PBS, the episode’s pairs join the two buffers; after many hands, θ_v is refit on the value buffer (regression) and θ_π is distilled from the policy buffer — closing the loop: *today’s average policies become tomorrow’s warm starts*. **At test time against a human, the identical loop runs minus the stores** — search-with-net-leaves *is* the player; training merely makes the nets worth searching with.

Listing 3: ReBeL’s training loop — the six steps above, in code. The same machinery runs at test time, minus the buffer writes.

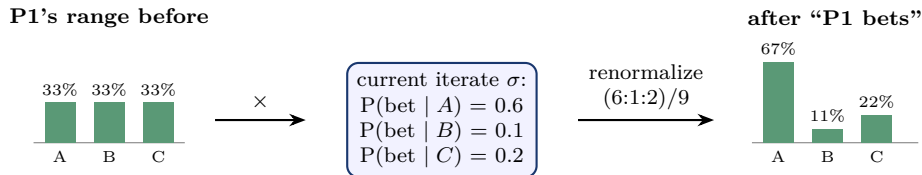
```
def rebel_episode(theta_v, theta_pi, v_buffer, pi_buffer):
    deal = sample_world()                # real cards for both seats
    beta = initial_pbs()                 # (1) blinds, uniform beliefs
    while not beta.is_terminal():
        G = depth_limited_subgame(beta)  # (2) SHAPE only, no values yet
        policies = run_cfr(G, T_cfr,      # (3) tabular CFR; EVERY iter:
                           warm_start=theta_pi) # leaf beliefs from current
                                                # sigma -> query V(.; theta_v)
        sigma_avg = average(policies)      # the converging object (vol. 1)
        v_buffer.add(beta,                 # (4a) value target: root values
                        value_of_root(G, sigma_avg)) # under the average
        pi_buffer.add(beta,                # (4b) policy target: the root's
                        sigma_avg.at_root()) # per-hand action distributions
        sigma = policies[randint(1, T_cfr)] # (5) play a RANDOM iterate
                                                # (why: close-up, Q2)
        hand = deal[beta.to_act()]
        a = sample(sigma[hand])            # (6) act with the real cards...
        beta = bayes_update(beta, sigma, a) # ...Bayes BOTH beliefs
    refit(theta_v, v_buffer)               # after many hands: regression
    refit(theta_pi, pi_buffer)             # + distillation -> warm starts
```

3.3 One solve under the microscope: a worked miniature

Steps 2–3 said “build the subgame, solve it with CFR”. To actually *see* that, shrink poker until the whole subgame fits in one picture: each player holds one of **three hand types** — *A* (the nuts), *B* (medium), *C* (air) — instead of 1326; the board is a flop, the pot is 6 chips, there is a single bet size (pot), and the depth limit is the end of this street. Both belief vectors at the root are uniform ($\frac{1}{3}, \frac{1}{3}, \frac{1}{3}$). Here is the *entire* subgame G_t that step 2 builds:

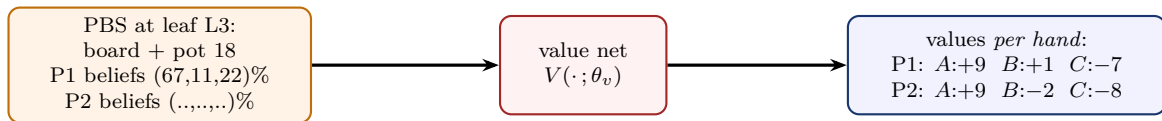


Two kinds of leaves, and the distinction is the whole method. **Fold terminals** (gray) need no intelligence: the pot goes to the survivor, arithmetic. **Depth leaves** (orange) are where the street ends and a turn card would be dealt — everything below them was cut off, so their value is a query to the value net, *at the leaf's own PBS*: same board and pot, but belief vectors that have been pushed through every action on the path. Concretely, here is that Bayes push for the root action “P1 bets”, with numbers — suppose the current CFR iterate has P1 betting 60% of its *A* hands, 10% of *B*, 20% of *C* (a value-bet/bluff mix):



This one picture is three lines of the listing at once: it is the `bayes_update` that produces the next root PBS, it is what happens *inside* the tree between a node and its child, and it is why the opponent must respect the bet — the bettor’s range really did get stronger (67% nuts), yet not pure (22% air), because the *solver* chose that bluff ratio.

Now “solve it” (step 3) is no mystery: run volume 1’s tabular CFR on this little tree — per-hand regret tables at the four decision nodes, sweeps $t = 1..T_{\text{cfr}}$ — with one twist at the orange leaves:



(numbers illustrative; what matters is the shape: beliefs in, one value per hand out)

And here is the subtle point that makes everything cohere: **the beliefs at a leaf change as CFR iterates**. Iteration 5’s σ bets a different mix than iteration 80’s, so the *same* leaf L3 is reached

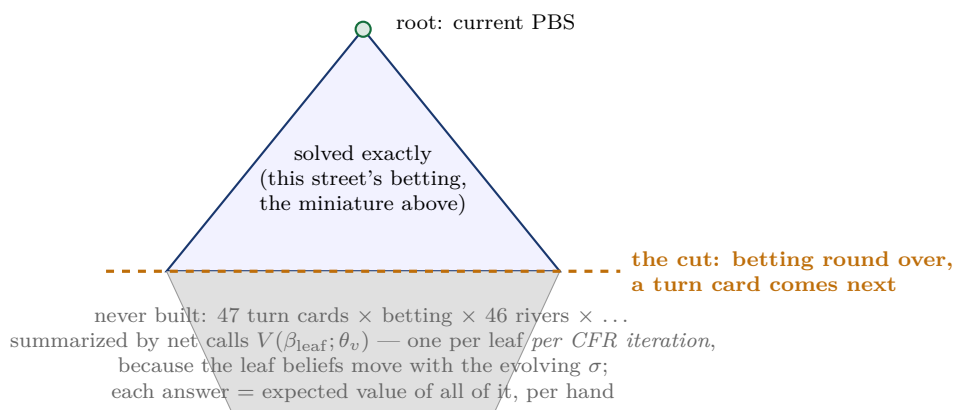
with *different* belief vectors, and the net must answer differently — a strong-range bet-call leaf is worth more to P1 than a bluff-heavy one. That is exactly why V takes the beliefs as input: one network must value *every possible range shape* at every public situation. Scaling back up to real HUNL changes only sizes — 3 hand types become 1326, one bet size becomes a menu, one street becomes the rest of the round. Nothing conceptual changes.

Intuition

Where Deep CFR and ReBeL put the neural network. Deep CFR’s nets *replace the regret tables* inside one global solve. ReBeL’s net replaces *the bottom of the search tree*: CFR itself stays tabular but only ever sees a small window of the game, and the net summarizes everything below the window. That is exactly AlphaZero’s division of labor, transplanted to imperfect information — which is why the paper’s title says “RL + Search”.

3.4 Two phrases from the listing, unpacked

“Depth-limited subgame” — where exactly is the cut, and why there? Picture the *full* game tree hanging below the current PBS: all betting sequences of this street, then ~ 47 possible turn cards *each* with all their betting sequences, then ~ 46 rivers each with theirs. The subgame keeps only the top slice — every betting sequence of the *current* street — and cuts exactly where the next card would be dealt, because that chance node is where the tree multiplies by 47 and becomes hopeless:



A chess engine does the same thing when it searches 12 plies and then calls its evaluation function instead of playing to checkmate. ReBeL’s “evaluation function” just answers a harder question — per hand and belief-aware — and is trained, not hand-written. The cut placement is the **dial between compute and trust**: cut deeper and you do more exact CFR work but lean on the net less; cut shallower and search is cheap but the net’s approximation error ($\varepsilon_{\text{approx}}$, volume 3) dominates. Cutting at the chance node is simply the sweet spot where each net call replaces the largest possible amount of tree. (And to be precise about *when* those calls happen: not at build time — a leaf has no value until beliefs reach it. The leaves are priced inside the solve, anew at every CFR iteration, as the previous section showed.)

“warm_start=theta_pi” — what exactly is being warmed? Start from what CFR’s internal state *is*: nothing but the regret tables $R[I]$ — the strategy is always derived from them by regret matching, $\sigma(a) \propto R^+(a)$. So “starting CFR” *means* choosing the initial contents of those tables. Cold start: all zeros $\Rightarrow$ iteration 1 plays uniform $\Rightarrow$ the first iterations are spent un-learning nonsense (volume 3 showed early iterates are garbage).

Watch out

Where does $\hat{\sigma}$ come from? Not from the value net — and it could not. ReBeL trains *two separate* networks. The value net θ_v answers “what is this leaf worth, per hand” and serves the depth-limit leaves. The policy net θ_π outputs *action probabilities directly*, per hand, and is trained by distillation: its targets are the average strategies that past subgame solves produced (just as AlphaZero’s policy head distills MCTS visit counts). One might hope to skip it and derive probabilities from values — a softmax over action values, say. Equilibrium play forbids this: *at a mixed equilibrium, every action in the mix has the same expected value* — that is what being willing to mix means. In a 70/30 bet/check spot, bet and check are worth exactly the same *to you*; the 70/30 is not there because betting is “better”, it is what keeps the *opponent* indifferent between her responses. The mixing ratio is a property of the equilibrium, invisible in your own values — probabilities must be solved for or distilled, never read off values.

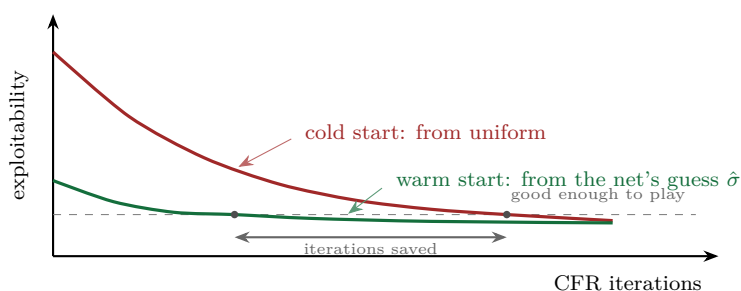
Now the trick, with numbers. Regret matching only looks at the *proportions* of the positive regrets. So if the policy net, shown this subgame’s PBS, predicts $\hat{\sigma} = (\text{bet } 70\%, \text{ check } 30\%)$ at some infoset, then initializing that infoset’s table to

$$R_{\text{init}} = w \cdot (0.7, 0.3) \quad (\text{any scale } w > 0)$$

makes regret matching reproduce exactly (70%, 30%) at iteration 1. Do this at every infoset of the subgame, and CFR *starts at the net’s guess* instead of at uniform.

The scale w is not cosmetic — it encodes **how much to trust the guess**, in units of “iterations’ worth of evidence”. Too small and one iteration of real regrets washes the guess out (nothing gained); too large and CFR becomes too stubborn to repair the net’s mistakes. Brown & Sandholm’s strategy-based warm start is the principled recipe for choosing w so the tables look *as if* the guess had been earned by genuinely played iterations — which is precisely why the convergence guarantee survives.

And the guess *is* repairable — that is the whole point of still running CFR. Suppose the net says “bet 70%” but in *this* exact subgame (these ranges, these sizes) betting is actually mediocre. Iteration 1 plays 70/30; the traversal reveals check’s values are higher; regret flows onto check; within a few dozen iterations σ has slid to, say, 40/60. The net saved the long journey from uniform; CFR still has the last word:



Last natural question: *if the net already knows a near-final strategy, why not skip CFR and just play $\hat{\sigma}$?* Because $\hat{\sigma}$ is an unverified generalization — the net has never seen *this exact* PBS, and playing its raw guess would be exploitable wherever the net happens to be wrong, with no guarantee left. Warm start is “trust, then verify”: the net proposes the usual route, the search drives it against today’s actual traffic.

Intuition

The two networks accelerate the two axes of the same solver. The **value net cuts the tree in space** (leaves replace everything below the chance node); the **policy net cuts the solve in time** (start near the answer instead of at uniform). Remove both and ReBeL degenerates into exact tabular CFR on the full game — correct and hopeless. The nets are not the player; they are compression of past search, spent to make the next search affordable.

3.5 What the value net actually sees and says

It is worth being concrete, because the input/output shape *is* the method. The input is the whole PBS: the public features (board cards, pot, bets) *plus both belief vectors* — ~ 1326 probabilities per player. The output is **not one number**: it is a **value per hand, for each player** (the *infostate values*). It has to be: the tabular CFR running above the leaves updates a regret *per hand* — “what is arriving at this leaf worth *if I hold AA? if I hold 76s?*” — so a single scalar would be useless to it. Equivalently: the net is a learned generalization of exactly the per-hand counterfactual values that DeepStack pioneered, with the beliefs as part of the input so one net covers every range shape.

The subgame ReBeL solves spans roughly the rest of the current betting round; wherever new public cards would be revealed, the tree stops and the net answers. Solving the subgame with T_{cfr} tabular CFR iterations queries the net thousands of times — this is why the net must be fast, and why its accuracy directly caps the quality of play (the $\varepsilon_{\text{approx}}$ floor of volume 3).

3.6 Close-up: the two questions everyone asks

Q1 — beliefs are computed from the players’ strategies. How can ReBeL know the opponent’s beliefs? It cannot, and it does not pretend to: like Pluribus’s ranges, the PBS is a statement about a *model*. ReBeL computes *all* beliefs — its own and its opponent’s — from the assumption that both players follow the policies its own procedure generates (this is the “common knowledge” assumption in the paper). Against an opponent who actually plays differently, the on-model logic of Pluribus’s Q2 applies verbatim: deviations are, within the model, unprofitable; early deviations can poison the beliefs; and the next question is precisely ReBeL’s elegant answer to that poisoning attack.

Q2 — after solving the subgame, why play the strategy of a randomly sampled CFR iteration rather than the final average? This is the strangest-looking line of the pseudo-code and the paper’s quiet masterstroke; it solves two problems at once.

- *Training consistency.* The value targets stored for the net are values of the subgame’s *average* policy — an average over all T_{cfr} iterations. If play (and hence the belief updates that produce the *next* PBS) always followed one fixed choice, the distribution of next-PBSs would not match the distribution the stored values implicitly assume, and the net would be trained on one distribution and queried on another — compounding off-distribution error down the game. Sampling the iteration uniformly makes “how beliefs are generated” agree, in expectation, with “how values were averaged”.
- *Safety.* Recall the hole in naive re-solving (Pluribus close-up, Q2): an opponent who knows how you re-solve can poison your beliefs. ReBeL’s theorem says that playing a uniformly sampled CFR iterate makes the depth-limited search *provably* low-exploitability — the same guarantee Libratus needed hand-built gadget games for, obtained instead by randomization. Intuition: the opponent cannot tailor an exploit to “the” strategy at the subgame root, because there is no single such strategy — they face the whole cloud of CFR iterates, and beating the cloud on average is exactly what CFR’s equilibrium guarantee forbids.

Intuition

A useful contrast triangle. **Libratus**: safety by *construction* (gadget subgames). **Pluribus**: safety *dropped* (six-max, empirical validation). **ReBeL**: safety by *randomization* (sample an iterate). All three are answers to the same question — “what stops an opponent from gaming my re-solver?” — and the three answers neatly span engineering, pragmatism, and theory.

Results. Superhuman heads-up no-limit hold’em with far less poker knowledge than DeepStack/Libratus: it beat top HUNL professional Dong Kim by **165 mbb/hand over 7 500 hands** (Kim was the best-performing human of the Libratus match). Also solid play in Liar’s Dice, showing generality. The experiments used a fixed menu of bet sizes for tractability; handling off-tree sizes by nested re-solving (as in Libratus/Pluribus) is compatible but orthogonal.

Watch out

The PBS is the whole trick — and the whole bill. The belief vector needs one probability per private state: 1326 in hold’em is fine; a game with 10^{10} private states is not. And beliefs are only well-defined if the policy they are computed from is *common knowledge* — ReBeL effectively “announces” its strategy, which is safe at equilibrium but subtle to reason about away from it. Porting ReBeL to a new game starts with the question: *can I even write down the belief vector?*

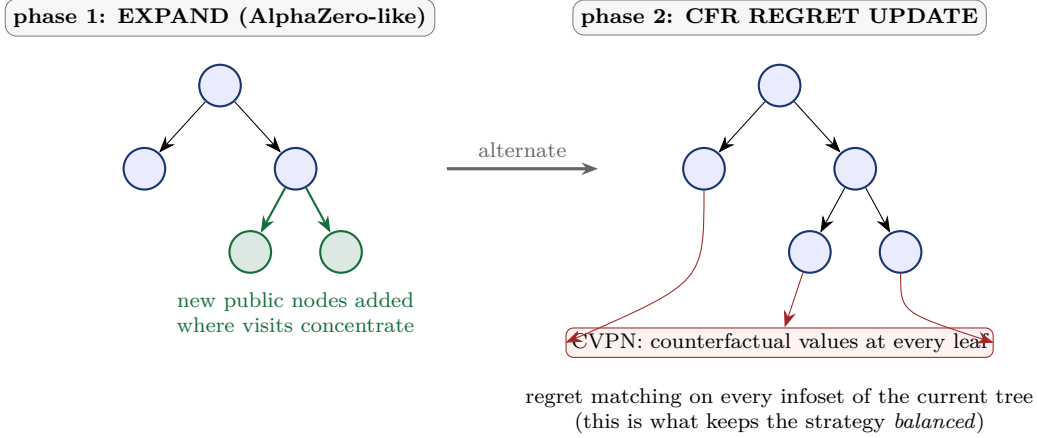
4 Player of Games: one sound algorithm for chess, Go and poker

Paper: Schmid et al., *Player of Games*, 2021; published in Science Advances 2023 under the name *Student of Games*. **One idea:** merge AlphaZero’s *growing* search tree with CFR’s *sound* handling of hidden information, so the same agent plays both kinds of games.

4.1 What “sound” buys you

AlphaZero’s MCTS is built for perfect information; applied naively to poker it converges to exploitable strategies (it has no concept of balancing a range). ReBeL is sound but poker-shaped. Player of Games (PoG) asks: what is the *minimal* unification? Its answer has two components:

- a **counterfactual value-and-policy network (CVPN)**: given a public state plus beliefs (like ReBeL’s PBS), it outputs *per-infoset counterfactual values* for both players and a policy prior. In a perfect-information game the beliefs are trivial and it degrades exactly to AlphaZero’s value+policy net.
- **growing-tree CFR (GT-CFR)**: the search keeps a small *public* tree and alternates two phases — *expand* the tree where the policy wants to visit (AlphaZero-style), then run *CFR regret updates* over the whole current tree with CVPN values at the leaves.



Listing 4: GT-CFR and sound self-play (schematic).

```
def gt_cfr(beliefs, cvpn, n_rounds):
    tree = PublicTree(root=beliefs)          # starts tiny
    for _ in range(n_rounds):
        for _ in range(E):                  # -- EXPAND --
            leaf = descend_by_policy_and_visits(tree)
            tree.add_children(leaf, prior=cvpn.policy(leaf))
        for _ in range(R):                  # -- REGRET UPDATE --
            cfr_iteration(tree, leaf_values=cvpn.values)
    return tree.root_policy(), tree.root_values()

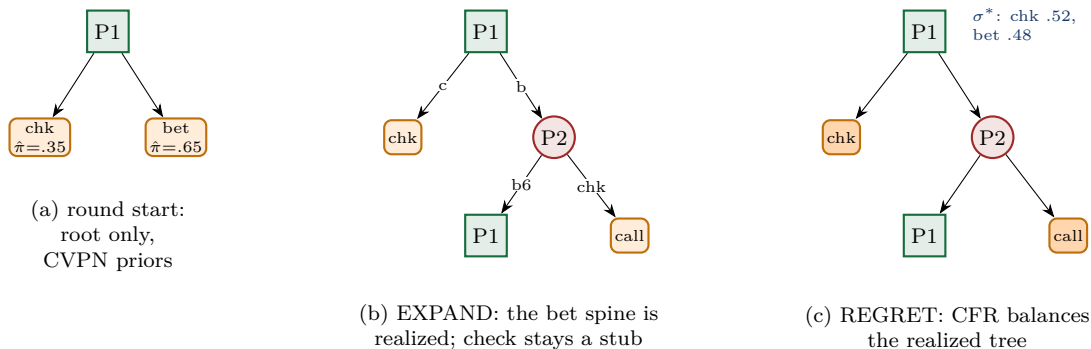
def sound_self_play(cvpn):
    while True:
        traj = play_game(act=lambda b: gt_cfr(b, cvpn, N))
        # every position where the net was queried at a leaf goes into a
        # QUERY BUFFER; a background process re-solves those positions
        # (recursively, with more search) to produce improved value targets
        train(cvpn, value_targets_from(query_buffer), policy_targets_from(traj))
```

Intuition

The query buffer is the conceptual heart of the training. The net is trained *at exactly the places where the search trusted it* — each leaf query becomes a promise to go back later, search deeper from that spot, and return with a better value to learn from. Compare our project’s stage-1 oracle: the principle “validate the function where you actually rely on it” is the same; PoG just turns it into the training signal itself.

4.2 One GT-CFR round under the microscope: growing a tree

The listing’s two inner loops — EXPAND then the CFR REGRET sweep — are clearest on the same miniature used for ReBeL: each player holds one of three hand types *A* (nuts), *B* (medium), *C* (air); flop pot 6; a single bet size; the street ends at the depth limit. What changes from ReBeL is *what gets built*. ReBeL constructed the *entire* subgame in one shot and solved all of it; GT-CFR starts from almost nothing and *grows* into the regions its policy prior cares about. Watch one round.



Round start (a) — the tree is a stub. The only realized node is the root: P1 to act, pot 6, uniform beliefs. Its two actions are not yet expanded into children; instead the CVPN, queried at the root PBS, attaches a *policy prior* $\hat{\pi}$ and a counterfactual value to each. Say $\hat{\pi} = (\text{check } .35, \text{bet } .65)$ — the net’s hunch that betting is the busier branch here.

EXPAND (b) — the AlphaZero move. To choose *where* to add a node, PoG descends by the PUCT score, exactly AlphaZero’s selection rule:

$$\text{score}(a) = \underbrace{Q(a)}_{\text{value so far}} + c \cdot \hat{\pi}(a) \cdot \frac{\sqrt{\sum_b N(b)}}{1 + N(a)}.$$

With every visit count N still zero the bonus is pure prior, so the *bet* branch is realized first: a P2 node appears, the CVPN is queried there for its children’s priors and values, and the next expansion descends again into P2’s highest-scoring reply. After E expansions only a thin spine exists — the lines the prior thought were live — while *check* remains an unrealized stub, valued wholesale by its CVPN counterfactual value. Nothing built the dense both-branches-to-the-bottom tree ReBeL paid for.

REGRET (c) — the CFR move. Now run the CFR sweeps over *whatever tree currently exists*. Every realized internal node carries per-hand regret tables; every leaf — a CVPN stub or a fold terminal — supplies per-hand values (the stub’s from the net, belief-aware, re-priced each sweep exactly as in ReBeL). Regret matching at the root slides P1’s *actual* play from the prior’s .65 bet toward whatever this tree’s values justify — here .48 — balancing the range so P2 stays indifferent. The prior proposed *where to look*; CFR decided *what to play*.

Alternate, and the tree thickens where it matters. The next round’s expansions use the freshly updated visits and values, so nodes accrete on the branches CFR keeps returning to while the perpetually-ignored lines stay stubs. Over many rounds the realized tree is a sparse skeleton hugging the relevant lines — the structure that lets the same code survive chess’s thirty-way branching, where ReBeL’s build-it-all step is a non-starter.

Intuition

Two solvers, one tree, divided by question. The expansion loop is AlphaZero asking “*which branch is worth a node?*” — visit- and prior-driven, argmax-flavoured. The regret loop is CFR asking “*what frequencies here are unexploitable?*” — mixing, regret-driven. They share the one growing public tree. Delete the expansion loop and you are back to ReBeL’s dense subgame; delete the regret loop and you are back to AlphaZero, exploitable the instant hidden information matters.

4.3 What a node of the search tree actually is

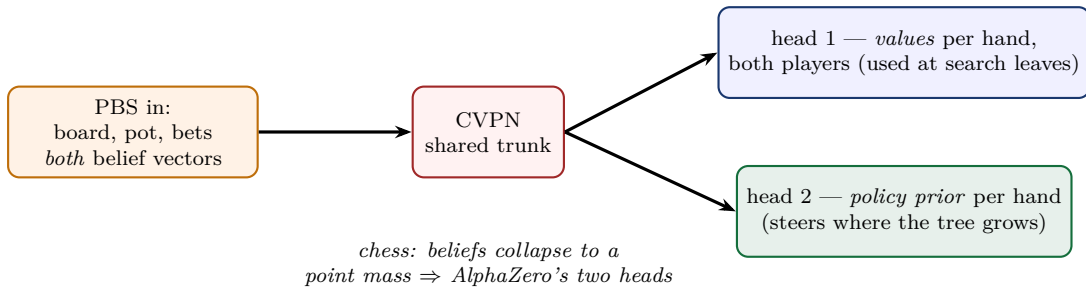
Same exercise as for ReBeL, because here too the data shapes *are* the method. A node of PoG’s tree is a *public* state **plus the players’ belief vectors** — a PBS in ReBeL’s vocabulary. The

CVPN takes that in and emits two things *per hand*: a counterfactual value (used when the node is a search leaf) and a policy prior (used to decide where to grow the tree, exactly like AlphaZero’s policy head guides PUCT). Now run the degeneration test: in chess, the “belief vectors” are trivial (one world state, probability 1), the public tree *is* the game tree, the counterfactual values collapse to ordinary state values — and PoG’s machinery reduces to AlphaZero’s value-plus-policy search. The imperfect-information generality costs nothing when there is nothing to be uncertain about. That is the precise sense in which it is “one algorithm for both kinds of games”.

The training loop runs two processes. *Actors* play games with a modest search budget, logging every leaf position where the CVPN was queried into the **query buffer**. A *trainer* pops queried positions and re-solves them with a *larger* budget, producing better values than the net originally returned — those become its regression targets (and the deeper solves may query new leaves, which re-enter the buffer with decaying probability: a self-improvement ladder in which the network forever chases its own deeper searches).

4.4 What the CVPN actually sees and says

As with ReBeL, the input/output shape *is* the method — with one revealing twist. ReBeL used *two separate* networks (θ_v for leaf values, θ_π for warm-start priors). PoG fuses them into **one network with two heads**, exactly AlphaZero’s value-plus-policy trunk:



Why fuse what ReBeL split? Both outputs are functions of the same PBS, so a shared trunk is natural — and, decisively, in PoG the policy head is *load-bearing*: it steers tree growth (the PUCT prior), not merely a warm start, so it must be as good as the value head and is worth training jointly. ReBeL kept its two apart because its signals are genuinely different — the value net learns by *regression* onto solve outputs, the policy net by *distillation* of average strategies — and its policy net is only an accelerator the search may overrule. Same two quantities; different division of labour.

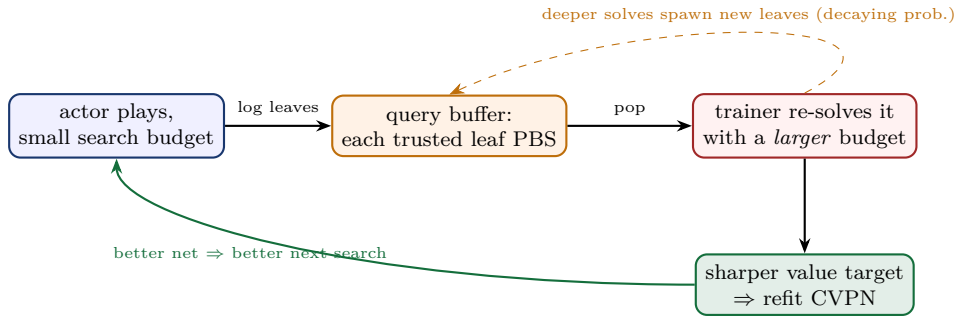
The degeneration test, in I/O terms. Set the game to chess: the belief vector is a point mass (one world state, probability 1), so the “per-hand” value vector collapses to a single scalar and the “per-hand” prior to a single move distribution — and the CVPN *is* AlphaZero’s (value, policy) net, bit for bit. Hidden-information generality costs nothing when the hand is public; that collapse is the whole content of “one algorithm”.

4.5 Two phrases unpacked

“Grows a sparse tree” — by what rule, exactly? The same PUCT selection AlphaZero uses, with one swap. Visits flow to actions scoring high on $Q(a) + c\hat{\pi}(a)\sqrt{\sum_b N(b)}/(1 + N(a))$ — high prior, high value so far, few visits spent. The swap: the $Q(a)$ feeding that score is *not* an MCTS average of rollout returns; it is the value of a under the *CFR solution of the current tree* — belief-aware, mixed, sound. Growth is steered by AlphaZero’s bandit arithmetic but fed by CFR’s values, which is why expanding the tree never quietly biases it toward a single deterministic line the way pure MCTS would.

“Sound” — a guarantee about what? Precisely this: with a bounded-error CVPN, GT-CFR’s exploitability is bounded and *shrinks as the tree grows*. That is the imperfect-information analogue

of “searching one ply deeper cannot hurt.” The contrast that gives it teeth: in imperfect information *plain MCTS has no such property* — more visits make its play more deterministic (Q2 below), which can make it *more* exploitable. “Sound” is exactly the guarantee MCTS forfeits the moment hidden information appears, and the one GT-CFR buys back by doing its searching with CFR.



The query buffer is what *earns* the bounded error. Every leaf the actor’s modest search trusted is logged; the trainer re-solves it with a larger budget, returns a sharper value, and the net regresses toward it — and those deeper solves query their *own* leaves, which re-enter the buffer with decaying probability. The network forever chases values produced by more search than it was allowed at play time: a ladder whose every rung is “be right where you were relied upon”.

Watch out

Soundness is conditional, and the bill is the belief vector. Two caveats travel together. First, the guarantee is *given* bounded value error — a CVPN that generalizes badly to an unseen PBS breaks the premise, not merely the play. Second, PoG inherits ReBeL’s entry fee: it must write down a belief vector per public state, so a game with astronomically many private states, or no clean notion of “public state”, is out of reach. “One algorithm for chess, Go and poker” is not “one algorithm for every game”.

4.6 Close-up: the two questions everyone asks

Q1 — ReBeL also searches over belief states. What is actually different? The *shape of the searched region*. At every decision, ReBeL builds the **full, dense** subgame down to its depth limit and solves all of it with tabular CFR — affordable in poker, where a betting round has a handful of public actions, but hopeless in games with large public branching or long horizons between chance events. PoG instead **grows a sparse tree** one node at a time, AlphaZero-style: the policy prior and visit counts decide which few branches deserve nodes at all, and the CFR phase keeps whatever tree currently exists balanced. Sparse growth is what lets the same code play chess (huge branching, no beliefs) and Scotland Yard (long imperfect-information horizons) — and it is why PoG, not ReBeL, is the one that generalized beyond poker.

Q2 — why does plain AlphaZero/MCTS fail at poker in the first place? Because MCTS is an *argmax engine*: as visits accumulate, it concentrates on the single best-looking action, and its play at a state becomes essentially deterministic. In perfect information that is exactly right — some move *is* best. In poker, determinism is a tell. A deterministic policy at a bluffing spot either always bluffs or never bluffs, and either way an observant opponent prints money (fold always / call always). Equilibria in imperfect information are *mixed* — calibrated frequencies, not best moves — and producing mixtures is precisely what regret matching does and tree-search argmax does not. GT-CFR is therefore not a hybrid for hybrid’s sake: the expansion phase brings MCTS’s scalability, the regret phase brings CFR’s balance, and each compensates the other’s failure mode.

Intuition

One sentence to keep: *AlphaZero asks “what is the best move here?”; CFR asks “what frequencies keep me unexploitable here?”* — and Player of Games is the observation that you can ask the first question about *where to look* while asking the second about *what to play*.

Results. One set of weights per game, same algorithm: expert-level chess and Go (below a specialized AlphaZero at equal compute — the price of generality), beats Slumbot (the strongest open heads-up poker bot) and is not exploited by a local best-response probe, and defeats the state of the art in Scotland Yard, a hide-and-seek board game with imperfect information.

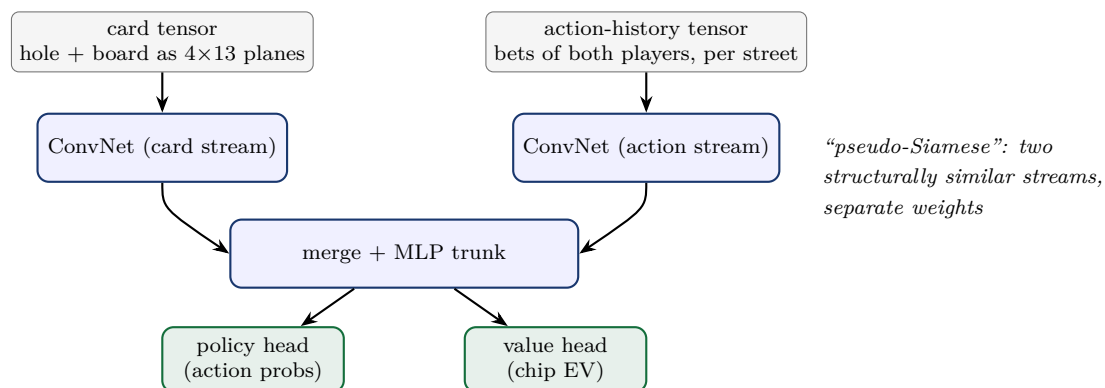
Watch out

Soundness here is a statement about the *search*, given good values: more search cannot make play more exploitable. It is not a global optimality proof, and in perfect-information games generality costs head-to-head strength versus a specialist. “One algorithm for everything” and “the best algorithm for each thing” remain different targets.

5 AlphaHoldem: what if we drop the game theory entirely?

Paper: Zhao, Yan, Li, Li, Xing, *AlphaHoldem: High-Performance Artificial Intelligence for Heads-Up No-Limit Poker via End-to-End Reinforcement Learning*, AAAI 2022. **One idea:** no CFR, no search, no beliefs — a single network maps the observation directly to an action, trained with a carefully stabilized PPO against a pool of past selves.

5.1 Architecture: two eyes, one brain

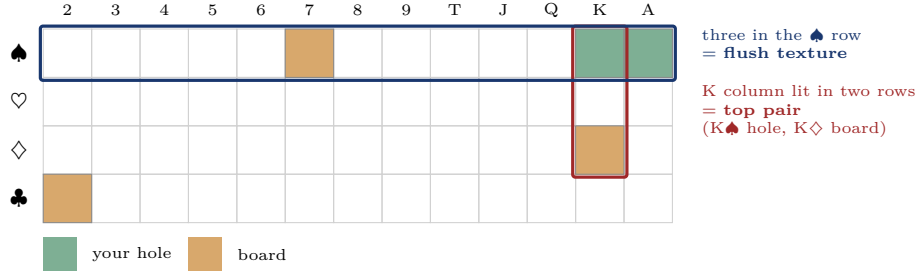


Note what is *absent*: no belief vector, no info set enumeration, no regret anything. The action history tensor plays the role our token sequence plays in this project — it is the only channel through which the network can infer what the opponent’s range looks like.

Why these particular encodings? The card stream stacks 4×13 binary planes (suit $\times$ rank), one group per street: hole cards, flop, turn, river. On that grid, poker structure is *geometric*: a straight draw is a run of adjacent *columns*, a flush draw is mass concentrated in one *row*, a paired board is two cells sharing a column — exactly the local patterns small convolutions detect cheaply. The action stream encodes, per betting round and per player, which action was taken and the normalized bet size, again as planes, so the same convolutional machinery picks up betting *patterns* (e.g. “raise–raise” textures) rather than treating the history as an opaque flat vector — the same lesson, in convolutional form, that pushed this project from a flat 816-dim input to a token sequence.

5.2 One observation under the microscope: the two tensors

Because “the data shapes are the method” here too, it pays to encode one concrete situation by hand. Hero holds $A\spadesuit K\spadesuit$; the flop is $K\heartsuit 7\spadesuit 2\clubsuit$. The card stream lays this on a 4×13 grid (suit $\times$ rank), lighting one cell per known card:



Read the grid the way a convolution does. The made pair is *two cells sharing a column* (K of two suits); the flush texture is *mass along one row* (three spades); a straight draw would be a *run of adjacent columns*; a paired board, two cells in one column. Every poker-relevant pattern is a *local* shape on this grid — precisely what small kernels detect cheaply, and precisely why the cards are planes and not a flat list. One group of planes per street (hole, flop, turn, river) keeps “when” alongside “what”.

The action stream is the second eye. It encodes, per street and per player, which action was taken and the normalized bet size, again as planes, so the same convolutional machinery reads betting *patterns* rather than an opaque flat history:

	P1	P2
pre	raise 2.5	call
flop	bet .5p	call
turn	check	check
river	bet 1p	—

same grid logic: a “raise-raise” texture becomes a local pattern the conv can name, not an opaque flat history — this is the implicit channel through which the net infers the opponent’s range

5.3 Making PPO survive poker

Vanilla PPO collapses on poker: rewards span ± 200 big blinds with brutal variance, and importance ratios explode on rare lines. AlphaHoldem’s **Trinal-Clip PPO** adds two clamps to the standard clipped loss:

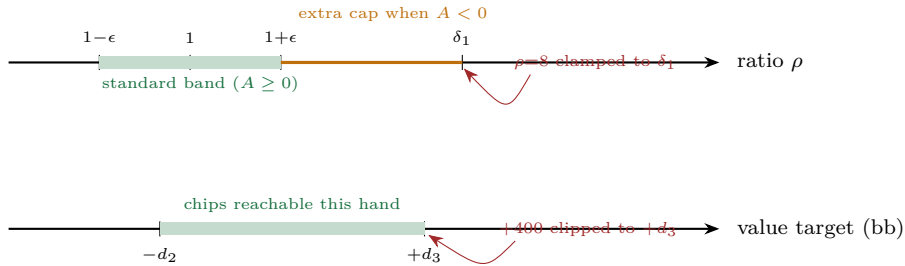
Listing 5: Trinal-Clip PPO (the three clips). delta1 caps the policy ratio when the advantage is negative; delta2/delta3 cap the value target at what is physically winnable/losable in the hand.

```
def trinal_clip_losses(ratio, adv, v_pred, v_target, eps, d1, d2, d3):
    # 1+2) standard PPO clip, PLUS an upper cap d1 (> 1+eps) on the ratio
    #      when adv < 0 -- a huge ratio times a negative advantage otherwise
    #      yields an unbounded, variance-exploding policy loss
    r = clip(ratio, 1 - eps, 1 + eps) if adv >= 0 else \
        clip(ratio, 1 - eps, d1)
    policy_loss = -min(ratio * adv, r * adv)
    # 3) value target clipped to the chips actually reachable this hand
    #      (d2 = max losable, d3 = max winnable -- pot/stack dependent)
    value_loss = (v_pred - clip(v_target, -d2, d3)) ** 2
    return policy_loss, value_loss
```

Numbers make the two failure modes concrete. Take a rare line — the new policy makes it $8\times$ likelier than the old, so the importance ratio is $\rho = 8$ — that turned out badly: advantage $A = -150$ bb.

Clip 1 (policy ratio, the δ_1 cap). The surrogate weights the update by $\rho \cdot A$. Left unbounded, that single rare transition contributes $8 \times (-150) = -1200$ to the objective — a gradient spike that swamps the hundreds of ordinary transitions sharing the batch. Standard PPO’s $[1-\epsilon, 1+\epsilon]$ band does not rescue the *negative*-advantage side; the extra upper cap δ_1 does, by refusing to let ρ count for more than δ_1 there. The heavy right tail of the ratio histogram is simply truncated.

Clips 2–3 (value target, the reachable band). A bootstrapped value target can overshoot what the hand can physically pay. If only 120 bb is live (the pot plus the stacks that can still go in), a regression target of $+400$ is an impossibility the critic should never chase. Clipping the target to $[-d_2, d_3] = [-120, +120]$ — max losable, max winnable *this* hand — keeps value learning inside what the chips allow.



Both clips encode the same physical fact: the chips on the table bound every learning target. The ratio cap bounds the policy update’s reach on a rare line; the value band bounds the critic’s target to what the pot can pay. Poker’s heavy tails are real money, not noise to average away — but they are finite, and that finiteness is exactly where it is safe to clip.

5.4 K-best self-play against strategy cycling

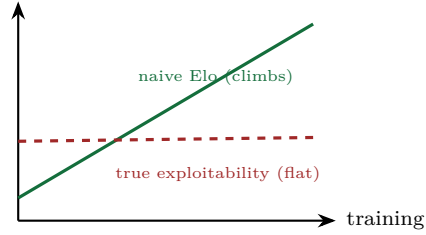
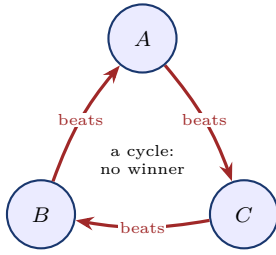
Naive self-play (always train against the latest self) famously *cycles* in poker: agent B beats A, C beats B, and C loses to A again — rock-paper-scissors in strategy space, no monotone progress. AlphaHoldem trains each new agent against the *K best historical versions* simultaneously, scored by an Elo-like rating; a new agent enters the pool only if it earns its place. (AlphaStar’s league is the same idea at larger scale.)

Listing 6: K-best self-play.

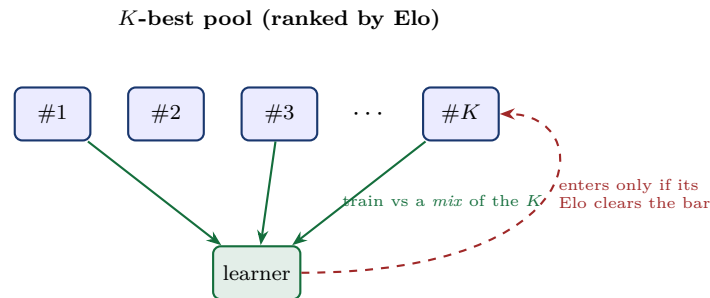
```
pool = [random_init()]
while training:
    learner = clone(best(pool))
    for _ in range(n_steps):
        opponent = sample(pool)           # mix over the K best
        rollouts = play(learner, opponent)
        update(learner, trinal_clip_ppo(rollouts))
    pool = top_k(pool + [learner], K, by=elo)  # survival of the fittest
```

5.5 The cycle under the microscope: rock–paper–scissors in strategy space

Q2 below names the disease; a picture makes it unforgettable. Three self-play checkpoints can chase each other in a directed cycle — *B* beats *A*, *C* beats *B*, and *A* beats *C* — so head-to-head “progress” is a carousel, not a climb. A naive rating, which only ever sees the latest pairwise result, marches upward the whole time:



The fix is to make “entering the pool” demand strength against a *spread* of past styles at once — a crude integral over the orbit rather than a single point on it. Each new learner trains against, and is rated against, the K best historical agents; it joins the pool only if its Elo clears the bar:



This is the same medicine AlphaStar’s league used at industrial scale, and — as Q2 argues — it is really CFR’s averaging theorem smuggled back in through the opponent distribution: you cannot carry an explicit average strategy, so you force the *opponents* to be an average instead.

5.6 Two phrases unpacked

“Pseudo-Siamese” — twins, but not identical. A true Siamese network runs *one* set of weights over two inputs of the *same* kind (two faces, say) to compare them. AlphaHoldem’s two streams share only the *shape* — a small ConvNet over a grid — not the *weights*, because their grids mean different things: a kernel that learned to read a flush draw as mass along a suit row would be nonsense slid across the action-history planes, whose axes are streets and players. “Pseudo” is the admission that the symmetry is architectural, not semantic — structurally twinned feature extractors, separately trained.

“End-to-end” — what was deleted, and what stands in. Lay the volume’s machinery beside AlphaHoldem and read off the subtractions:

- **ranges / beliefs — gone.** No PBS, no info set enumeration. The action-history ConvNet must *implicitly* infer “what does the opponent’s range look like” from the betting planes alone.
- **the search tree — gone.** No subgame, no CFR-at-play-time. The single forward pass *is* the decision (2.9 ms, Results below).
- **regret and averaging — gone.** No explicit average strategy. The PPO gradient plus the K-best pool stand in for CFR’s averaging theorem, as the cycle picture above makes literal.

And yet poker knowledge does not vanish — it relocates. The value clip’s reachable band, the 4×13 card geometry, the per-street action planes, the K-best schedule: each is domain-specific scaffolding doing work that CFR’s structure used to do for free. “End-to-end” describes the unbroken gradient path from observation to action, not an absence of poker priors.

Watch out

“End-to-end” hides its inductive biases in the plumbing. It is tempting to read AlphaHoldem as “just let SGD figure out poker”. But the encodings and the clips *are* the priors that make the gradient well-behaved; swap the card grid for a flat vector, or drop the reachable-chips value clip, and the same PPO diverges. The lesson for this project is the mirror image of its own history: the move from a flat 816-dim input to a token sequence was exactly this kind of inductive-bias engineering, and it mattered every bit as much as the learning rule.

5.7 Close-up: the two questions everyone asks

Q1 — no search, no ranges: where does balance (bluffing) come from, if anywhere?

Only from gradient pressure. If the current policy is predictable — bets only its strong hands — then *provided some opponent in the K-best pool punishes that pattern*, those rollouts lose money and the policy gradient pushes toward mixing. Balance can emerge, but it is an *equilibrium of the pool*, not of the game: nothing certifies that the pool contains the punisher for every leak. Contrast the other systems in this volume: Pluribus *computes* balance per spot (Trick 2, the whole-range solve); AlphaHoldem *hopes SGD finds it*. This is the precise content of the “no certificate” trade — and why follow-up work exploiting pure-RL poker agents keeps finding leaks that self-play ratings never showed.

Q2 — why does naive self-play cycle, and why does a K-best pool help? Volume 3 proved the deep reason: in zero-sum games, the *current* strategies of self-play never converge — they orbit the equilibrium forever (the RPS table showed a 73%-Paper current strategy mid-convergence); only the *average* converges. CFR is built around that fact: it carries the average explicitly. A vanilla RL agent keeps only a current policy, so it *lives* the orbit instead of averaging it: B learns to beat A, C learns to beat B, and C loses to A — rock-paper-scissors in strategy space, with ratings that go up forever while true strength goes nowhere. Training against the *K* best *historical* agents at once is a crude integral over the orbit: to enter the pool you must do well against a *spread* of past play styles simultaneously, which damps the cycle — a poor man’s strategy averaging. (AlphaStar’s league is the same medicine at industrial scale.) Seen this way, K-best self-play is not an RL hack; it is CFR’s averaging theorem, smuggled back in through the opponent distribution.

Intuition

The two clips of Trinal-Clip PPO have the same shape as this project’s regret-scale reasoning: bound every learning target by what is *physically reachable in the hand* ($\pm$ the committed chips). Heavy tails in poker are not noise to average away — they are impossible targets to refuse.

Results. Over 100 000 evaluation hands, AlphaHoldem beats Slumbot and a DeepStack re-implementation, after **3 days of training on one 8-GPU server**, and then decides in **2.9 ms** on a single GPU — roughly a thousand times faster than search-based systems.

Watch out

No theory, no certificate. Nothing bounds AlphaHoldem’s exploitability; “beats Slumbot” does not imply “cannot be exploited by a strategy that hunts for its leaks” — and follow-up work has indeed found pure-RL poker agents exploitable. This is the precise trade this corner of the design-space map makes: spectacular speed and simplicity, no safety net. Our project’s self-play collapses (the 1M-step incoherent policy) were a small taste of the same phenomenon — and our stage-1 Nash oracle is exactly the kind of certificate AlphaHoldem lacks.

6 Side by side, and what this project takes from each

	Deep CFR	Pluribus	ReBeL	PoG/SoG	AlphaHoldem
year	2019	2019	2020	2021	2022
core engine	CFR	CFR	CFR+RL	CFR+MCTS	PPO
neural nets	value+policy	none	value+policy	CVPN	policy+value
search at play time	no	yes	yes	yes	no
belief states needed	no	no	yes	yes	no
abstraction needed	no	yes	no	no	no
players	2	6	2	2	2
theoretical target	Nash	none (empirical)	Nash	sound search	none
training hardware	GPUs	64 CPU cores, 8 days	many GPUs	TPUs	8 GPUs, 3 days
decision latency	ms	seconds	seconds	seconds	2.9 ms

Intuition

Choosing for this project. PokerTrainer sits in the Deep CFR cell on purpose: no abstraction to hand-craft (the token transformer reads raw infosets), no play-time search infrastructure (one network ships in the Flutter inspector and the C ABI), and a real theoretical target (Nash) that gives us measurable progress — which the stage-1 push/fold oracle exploits directly. The map’s other corners tell us what we are giving up: Pluribus-style re-solving would buy precision at the cost of an online solver; ReBeL/PoG belief machinery would buy soundness-at-depth at the cost of 1326-dim belief plumbing; AlphaHoldem’s pure RL would buy simplicity at the cost of the very guarantees we built the oracle to enforce.

Concrete techniques worth borrowing here, in increasing order of effort:

1. **K-best opponent pools (AlphaHoldem)** — our evaluation matches already revealed self-play drift; rating checkpoints with Elo and evaluating new ones against the K best is cheap and directly transplantable.
2. **Reward/value clipping by reachable chips (AlphaHoldem)** — the same physical bounds ($\pm$ effective stack) we used to reason about regret scale apply to any value-learning component we add.
3. **Continuation-strategy leaf evaluation (Pluribus)** — if we ever add depth limits beyond the current crude bootstrap (flagged in `cfr_coro.py`), the 4-continuation construction is the principled upgrade.
4. **Query-buffer-style targeted validation (PoG)** — extend the stage-1 oracle idea: log the infosets where the AdvNet’s regrets most influenced traversals, and audit/validate exactly there.

7 Further reading

- Brown, Sandholm (2019). *Superhuman AI for multiplayer poker* (Pluribus). Science.
- Brown, Sandholm (2018). *Depth-Limited Solving for Imperfect-Information Games* — the continuation-strategies construction Pluribus builds on.
- Brown, Bakhtin, Lerer, Gong (2020). *Combining Deep Reinforcement Learning and Search for Imperfect-Information Games* (ReBeL). NeurIPS.
- Schmid et al. (2021/2023). *Player of Games / Student of Games: A unified learning algorithm for both perfect and imperfect information games*. Science Advances.

- Zhao, Yan, Li, Li, Xing (2022). *AlphaHoldem: High-Performance Artificial Intelligence for Heads-Up No-Limit Poker via End-to-End Reinforcement Learning*. AAAI.
- Moravčík et al. (2017). *DeepStack* — the origin of learned counterfactual values at depth limits, ancestor of both ReBeL and PoG.
- Companion volume: *Deep CFR, explained* ([docs/deep_cfr_tutorial/](#)) — regret matching, CFR, MCCFR, Deep CFR, and this project’s curriculum.